AWS ARCHITECT ASSOCIATE • PROJECT WORKBOOK

AWS VPC Peering

Connecting Heterogeneous Virtual Private Clouds & Proving Private Network Reachability

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01 Architecture Overview & Resource Inventory

The deployed architecture establishes bidirectional, non-transitive private connectivity between two non-overlapping Virtual Private Clouds: **VPC A (10.10.0.0/16)** and **VPC B (10.20.0.0/16)** in AWS Region **ap-south-1 (Mumbai)**.

> **Key Security Characteristic:** Neither `ewb-app-a` nor `ewb-app-b` possesses a public IPv4 address or direct internet routing. All remote administration is mediated through `ewb-bastion` in VPC A's public subnet using OpenSSH `ProxyJump`.

Resource Inventory Table

Resource Name	Logical Role	AWS Resource ID	Subnet / VPC	IPv4 / CIDR
ewb-vpc-a	Requester VPC	`vpc-084a10c21275c9423`	N/A	10.10.0.0/16
ewb-vpc-b	Accepter VPC	`vpc-0d5be64741a5ab8b9`	N/A	10.20.0.0/16
ewb-a-public	Public Subnet	`subnet-074a1bdc28f74eb7b`	ewb-vpc-a	10.10.1.0/24
ewb-a-private	Private Subnet	`subnet-0739012e0e05001b3`	ewb-vpc-a	10.10.2.0/24
ewb-b-private	Private Subnet	`subnet-083532ce04867e0d8`	ewb-vpc-b	10.20.1.0/24
ewb-igw-a	Internet Gateway	`igw-0852644c6bf4be208`	ewb-vpc-a	N/A
ewb-a-b-peer	VPC Peering Connection	`pcx-078073f607a5a75f0`	VPC A ↔ VPC B	Active (pcx)
ewb-bastion	Bastion Jump Host	`i-006304f40baed16dc`	ewb-a-public	10.10.1.10 (Public: 52.66.105.166)
ewb-app-a	Workload A Server	`i-0c4aabcc3f387bde1`	ewb-a-private	10.10.2.10 (No Public IP)
ewb-app-b	Workload B Server	`i-0e096e913d95c559a`	ewb-b-private	10.20.1.10 (No Public IP)

02 Route Table Configurations & Subnet Associations

To enable packet flow between instances and across the peering link, three distinct route tables are configured with explicit subnet associations:

Route Table Name	Associated Subnet	Destination CIDR	Target	Routing Purpose
ewb-rt-a-public	`ewb-a-public`	10.10.0.0/16	`local`	Local VPC A intra-communication
		0.0.0.0/0	`igw-0852644c6bf4be208`	Internet gateway route for Bastion
		10.20.0.0/16	`pcx-078073f607a5a75f0`	Enables Bastion to reach App-B for SSH
ewb-rt-a-private	`ewb-a-private`	10.10.0.0/16	`local`	Local VPC A intra-communication
		10.20.0.0/16	`pcx-078073f607a5a75f0`	Forward path: App-A to VPC B workload
ewb-rt-b-private	`ewb-b-private`	10.20.0.0/16	`local`	Local VPC B intra-communication
		10.10.0.0/16	`pcx-078073f607a5a75f0`	Return path: VPC B workload back to VPC A

Longest Prefix Match (LPM): In `ewb-rt-a-public`, traffic toward `10.20.1.10` evaluates `10.20.0.0/16` (more specific) over the default route `0.0.0.0/0`, routing SSH ProxyJump traffic directly through the peering connection instead of blackholing it.

03 Security Groups & Least-Privilege Access Control

Inbound firewall rules adhere strictly to least-privilege principles by restricting access to /32 host IPs rather than open CIDR ranges.

Security Group	VPC Scope	Type / Protocol	Port Range	Permitted Source	Security Rationale
ewb-sg-bastion	ewb-vpc-a	SSH / TCP	22	`122.170.198.164/32`	Restricts administrative ingress to the student's authorized IP
ewb-sg-app-a	ewb-vpc-a	SSH / TCP	22	`10.10.1.10/32`	Allows SSH solely from Bastion private IP
		Custom ICMP - IPv4	Echo Request (8/0)	`10.20.1.10/32`	Allows ICMP ping probes only from App-B
ewb-sg-app-b	ewb-vpc-b	SSH / TCP	22	`10.10.1.10/32`	Allows SSH solely from Bastion private IP via ProxyJump
		Custom ICMP - IPv4	Echo Request (8/0)	`10.10.2.10/32`	Allows ICMP ping probes only from App-A

Local SSH Configuration File (~/.ssh/config)

```
# Local OpenSSH Client Configuration Host ewb-bastion HostName 52.66.105.166 User ec2-user IdentityFile ~/.ssh/ewb-lab.pem IdentitiesOnly yes Host ewb-app-a HostName 10.10.2.10 User ec2-user IdentityFile ~/.ssh/ewb-lab.pem IdentitiesOnly yes ProxyJump ewb-bastion Host ewb-app-b HostName 10.20.1.10 User ec2-user IdentityFile ~/.ssh/ewb-lab.pem IdentitiesOnly yes ProxyJump ewb-bastion
```

04 Verification & Test Results

Test 01: Baseline Failure Test (Before Peering Routes)

Ping from `ewb-app-a` to `10.20.1.10` prior to adding peering routes proves complete network isolation.

```
[ec2-user@ip-10-10-2-10 ~]$ hostname -I 10.10.2.10 [ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. --- 10.20.1.10 ping statistics --- 4 packets transmitted, 0 received, 100% packet loss, time 3072ms
```

Test 02: Forward Path Verification (App-A → App-B)

After peering activation and route additions, ICMP Echo requests route successfully through `pcx`.

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. 64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.982 ms 64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.841 ms 64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.812 ms 64 bytes from 10.20.1.10:
```

```
icmp_seq=4 ttl=127 time=0.825 ms --- 10.20.1.10 ping statistics --- 4 packets transmitted, 4 received,
0% packet loss, time 3045ms rtt min/avg/max/mdev = 0.812/0.865/0.982/0.068 ms
```

Test 03: Reverse Path Verification (App-B → App-A)

Proves the return route in `ewb-rt-b-private` is active and symmetric.

```
[ec2-user@ip-10-20-1-10 ~]$ hostname -I 10.20.1.10 [ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10 PING
10.10.2.10 (10.10.2.10) 56(84) bytes of data. 64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.892 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.834 ms 64 bytes from 10.10.2.10: icmp_seq=3 ttl=127
time=0.819 ms 64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.848 ms --- 10.10.2.10 ping statistics
--- 4 packets transmitted, 4 received, 0% packet loss, time 3051ms rtt min/avg/max/mdev =
0.819/0.848/0.892/0.028 ms
```

Test 04: Private SSH ProxyJump Validation

Executing `ssh ewb-app-b` from local workstation hops through Bastion into the private host.

```
user@laptop:~$ ssh ewb-app-b Last login: 22 October 2026 from 10.10.1.10 , #_ ~\_ ####_ Amazon Linux
2023 ~~ \ _-### AL2023 AMI ~~ / # ~~ /vv~ https://aws.amazon.com/linux/amazon-linux-2023/ ~~ [ec2-
user@ip-10-20-1-10 ~]$ # Successfully authenticated on private App-B
```

05 Controlled Failure & Recovery Analysis

To definitively prove that the Security Group firewall is actively filtering traffic (and that ping success was not an artifact of an overly permissive rule), a controlled failure was executed by temporarily dropping the ICMP rule.

> **Fault Injection Procedure:** The inbound ICMP Echo Request rule (Type 8, Code 0 from `10.10.2.10/32`) was deleted from Security Group `ewb-sg-app-b` while keeping the SSH rule intact.

Phase A: Failure Observation (Forward Ping from App-A)

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. --- 10.20.1.10 ping statistics --- 4 packets transmitted, 0 received, 100% packet loss, time 3088ms
```

Technical Analysis: Although the VPC route table still successfully routed packets to `pcx` and delivered them to VPC B, the AWS hypervisor firewall (Security Group) dropped the incoming ICMP Echo Request at the virtual network interface of `ewb-app-b`. Note that SSH connectivity from Bastion remained fully operational during this period, proving port-level granularity.

Phase B: Recovery Verification

The exact ICMP rule was re-added to `ewb-sg-app-b`, instantly restoring connectivity:

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. 64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.887 ms 64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.829 ms --- 10.20.1.10 ping statistics --- 4 packets transmitted, 4 received, 0% packet loss, time 3039ms
```

06 Technical Defense & Viva Q&A

Q1: Why is the return route needed?
Answer: Network communication is bidirectional. Even if the requester VPC has a route sending packets across the peering connection, the destination host's VPC router needs an explicit route table entry for the return destination CIDR (`10.10.0.0/16 -> pcx`). Without it, return packets (such as ICMP Echo Replies or TCP SYN-ACKs) are dropped by the VPC router.

Q2: Why is the App-B SSH source the bastion private IP?
Answer: When using SSH ProxyJump, the SSH client establishes a hopped tunnel. The TCP connection to App-B is initiated by the Bastion's kernel using its private ENI IP (`10.10.1.10`). From App-B's network standpoint, the incoming TCP connection originates from `10.10.1.10`, not the student's laptop.

Q3: Why does Active peering not guarantee a successful ping?
Answer: Peering status indicates only that the tunnel is provisioned. Reachability requires three additional layers: (1) route tables on both subnets pointing to the peer, (2) Security Groups permitting inbound ICMP traffic, and (3) Network ACLs allowing inbound/outbound packets.

Q4: Can VPC B use VPC A's internet gateway over peering?
Answer: No. AWS VPC Peering is non-transitive. Traffic entering a VPC via a peering connection cannot be routed out through that VPC's Internet Gateway or NAT Gateway.

07 Teardown Evidence & Key Learnings

Following project completion and evidence capture, all cloud resources were terminated in reverse dependency order (EC2 instances terminated → Peering connection deleted → Route tables and subnets disassociated → IGW detached → VPCs deleted).

> **Key Takeaway:** Designing multi-VPC networks in AWS requires synchronized alignment of Layer 3 IP routing (Route Tables), Layer 4 firewalls (Security Groups), and connection topologies. VPC Peering provides high-throughput, low-latency private connectivity, but requires strict understanding of non-transitivity and explicit bidirectional route definitions.